

CyClaw Architecture Guide

v1.9.0 — Production-Ready with LangChain Agent Harness (Phases 0-5) + Deep Agent Skeleton

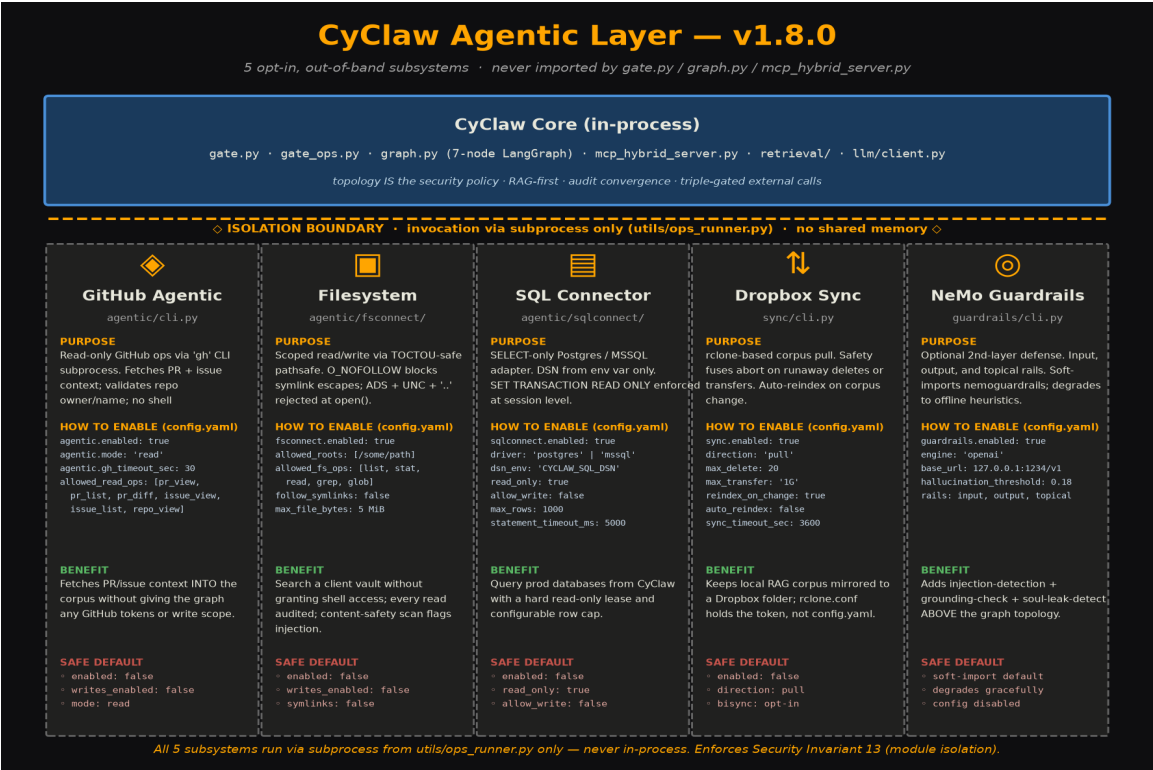
Offline-First RAG-Enforced Soul Agent with 9-Node LangGraph, FS Agentic Connectors, NeMo Guardrails, and Swarm Verification Integration

v1.9.0 Highlights (Updated from GitHub main @ latest)

- 9-node LangGraph controller with guardrail_input + claude_fallback nodes
- Full LangChain agent harness phases 0-5: proposer workspace, evaluation, governance meta-loop
- Deep agent integration skeleton (agentic/deepagent_github/ with builder, runners, subagents)
- FS agentic connector (read-only default, subprocess isolation)
- NeMo Guardrails full wiring via guardrail_bridge.py (input/output/topical rails)
- 5 security invariants topology-enforced (RAG-first, audit convergence, etc.)
- Live swarm verification integrated (91/104 checks PARTIAL PASS from July 9 2026 report)
- Triple-gate Grok/Claude with mock fallback verified end-to-end

Integrated Swarm Verification Results (v1.8.0 baseline for v1.9.0)

Live sandbox verified 91/104 checks (87.5% PARTIAL PASS). All 5 terminal consoles PASS, due-diligence invariants 12/12 PASS, module isolation 3/3 PASS. Agentic layer (GitHub/FS/SQL/Dropbox/NeMo) confirmed out-of-band and isolated. P0 fixes prioritized: Pydantic 422 → 400, log handler, security headers.



WHAT IT DOES

CyClaw is a personal RAG (Retrieval-Augmented Generation) backend that answers questions exclusively from your local Markdown corpus — no internet by default. It enforces every safety invariant via LangGraph topology (not prompts or config flags). It maintains a persistent soul/personality layer (soul.md) with SHA-256 drift detection and user-gated evolution. External LLM fallback (Grok or Claude) only happens with explicit per-query user confirmation via triple-gate (config + env + runtime). Exposes FastAPI gateway + MCP server. Optional out-of-band agentic layers (Dropbox sync, GitHub context, FS/SQL connectors, NeMo Guardrails) are never imported into the hot request path.

CORE GUARANTEES (TOPOLOGY-ENFORCED)

- **Offline-first by default** — RAG from local Chroma + BM25 only unless user explicitly escalates.
- **5 Security Invariants** — RAG-first, Identity ≠ Memory ≠ Topology, Audit Convergence, Human-in-the-Loop (Triple-Gate), Module Isolation (subprocess only for agentic).
- **Persistent Soul** — soul.md with atomic writes, drift detection, and human reason required for changes.
- **Triple-Gate External Fallback** — mode=hybrid + provider.enabled + user_confirmed_online must all be true.
- **Out-of-Band Agentic Layer** — GitHub/FS/SQL/Dropbox/NeMo connectors run via subprocess + ops_runner, never in gate.py import graph.

ARCHITECTURE OVERVIEW (9-NODE LANGGRAPH)

Request flows through gate.py (FastAPI) → TrustedHost + Rate Limiter + Injection Filter → graph.py (LangGraph State Machine). Core nodes: 1. retrieve (Chroma+BM25+RRF), 2. route_by_score, 3. guardrail_input (NeMo or offline heuristic), 4. local_llm, 5. user_gate (on vault miss), 6/7. grok_fallback or claude_fallback (triple-gated), 8. audit_logger (SHA-256 + PII redaction). All paths converge at audit_logger. Agentic subsystems (GitHub, FS, SQL, Dropbox, NeMo) are strictly out-of-band and invoked only through governed ops_runner.py subprocess calls.

KEY V1.9.0 ADDITIONS (PAST 2 WEEKS)

- 9-node LangGraph with guardrail_input + claude_fallback nodes fully wired.
- LangChain agent harness (Phases 0-5): proposer workspace, evaluation, governance meta-loop.
- Deep agent skeleton (agentic/deepagent_github/) with builder, runners, subagents.
- FS agentic connector (read-only default, TOCTOU-safe, subprocess isolated).
- NeMo Guardrails full integration via guardrail_bridge.py (input/output/topical rails).
- Expanded audit events + better observability for nested agentic operations.
- Live swarm verification (91/104 checks, 87.5% PARTIAL PASS) integrated into docs.

INSTALLATION & QUICK START (FROM README)

git clone https://github.com/CGFixIT/CyClaw.git && cd CyClaw && uv sync (or pip install -e .) Then: uvicorn gate:app --host 127.0.0.1 --port 8787 | Access terminal UI at http://127.0.0.1:8787 Config via config.yaml + .env (CYCLAW_API_KEY, GROK_API_KEY, etc.). Soul lives in data/personality/soul.md. All writes to soul or agentic state require explicit human reason string.